MILITARY SPECIFICATION
CONTROL SYSTEM, REMOTE, AN/FSQ-98(A)(V)

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a reliable, all solid-state Remote Control System, AN/FSQ-98(A)(V), hereinafter referred to as the RCS, to be used in conjunction with the AN/FRT-83(V), AN/FRT-84(V), and AN/FRT-85(V) independent side band (ISB) high frequency radio transmitters. The RCS employs serial digital data exchange to control and monitor from one up to, and including, 10 AN/FRT radio transmitters arranged in any combination or system configuration. Each of 10 AN/FRT radio transmitters used with the RCS may be individually controlled and monitored from a remote location.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic, Interior Communications And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-E-21981	Electronic Type Designations, Identification Plates And Markings, Requirements For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-188	Military Communication System Technical Standards
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-781	Reliability Design Qualification Production And Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1364	Preferred General Purpose Electronic Test Equipment
MIL-STD-1378	Military Standard Requirements For Employing Standard Hardware Program Modules

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA-RS-310-C-77 Racks, Panels And Associated Equipment

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The RCS shall be in accordance with MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified (see 6.2), the contractor shall furnish two sample units for first article inspection and approval (see 4.3 and 6.3). The sample units shall be suitable for complete evaluation of mechanical and electrical form, design, and performance. The sample units shall be of final mechanical and electrical form, and employ standard or nonstandard (when approved by the procuring activity) parts to be completely representative of the final equipment.

3.1.2 Equipment conditioning. An equipment conditioning test shall be an integral part of the first article or quality conformance inspection and shall provide full details of the test setup, functional checkout procedures, and data logging techniques when requested by the Government.

3.2 General design. General design shall be as specified in 3.2.1 through 3.2.19. The RCS shall contain a minimum parts count consistent with the performance specified herein and shall contain all of the design characteristics required to achieve a mean-time-between-failure (MTBF) of at least 2500 hours.

3.2.1 Selection of parts. Selection of parts should ensure maximum commonality of piece parts supplied under previous contracts.

3.2.1.1 Government-furnished equipment (GFE). GFE (see 6.7) provided for use on any contract resulting from this specification, may include nonstandard parts. Nevertheless, the equipment furnished in accordance with this specification shall employ standard pieces or approved non-standard parts.

3.2.1.2 Electron tubes. Electron tubes shall not be used in the design of the RCS to be provided in accordance with the requirements of this specification.

3.2.1.3 Semiconductor devices. Wherever possible semiconductor devices shall be selected from Government approved types and shall be silicon.

3.2.1.4 Encapsulation. Plastic encapsulated transistors and microelectronic devices shall not be used in the design of the RCS unless specifically approved in writing by the procuring activity.

3.2.2 Enclosure. Proper access for each unit shall be provided for normal repair and maintenance. Enclosure rigidity shall be sufficient to survive the shock tests specified in 4.5.3 without deforming. The enclosure surface shall be finished with enamel conforming to MIL-E-15090, Type III, Class 2.

3.2.3 Shock. The RCS shall withstand shock tests specified in 4.5.3 without performance degradation or malfunction.

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3.2.4 Nomenclature. Nomenclature shall be requested in accordance with MIL-E-21981.

3.2.5 Accessibility. The RCS shall be so constructed that parts, terminals, and wiring, are accessible for circuit checking, adjustment, maintenance, repair and replacement with a minimum disturbance to other parts and wiring, and with use of the minimum number of special tools (see 6.4). Special hardware tools needed to maintain the RCS shall be provided with the RCS. Parts shall be mounted so that their identification markings will be readily visible with minimum disassembly of the RCS. Requirements for special tools shall be in accordance with MIL-STD-454, Requirement 63.

3.2.6 Serial numbers. All major components of the RCS shall have serial numbers.

3.2.7 Insulating compounds. Compounds which soften and flow at any temperature less than 75°Celsius (C) shall not be used unless contained in such a manner that the compound will not flow from the container during service use. Compounds which crack at any temperature above -62°C shall not be used.

3.2.8 Electromagnetic compatibility characteristics. The requirements of Class A3 of MIL-STD-461 shall apply except that the radiated susceptibility test limits over the frequency range 10 kilohertz (kHz) to 400 megahertz (MHz) shall be 0.01 watts per square centimeter.

3.2.9 Modular construction. Modular unit construction shall be used and MIL-STD-1378 shall be considered in the design of the RCS.

3.2.10 Environmental conditions. The RCS shall be capable of normal operation, with performance as specified in 3.4.1, under the range of environmental conditions specified in 3.2.10.1 and 3.2.10.2.

3.2.10.1 Operating conditions. Operating conditions shall be as specified in a through c:

a. Ambient temperature: 0°C to 50°C, at sea level
b. Humidity: 0 percent to 95 percent relative humidity (RH)
c. Elevation: Sea level to 10,000 feet (ft), 20°C at 50 percent RH

3.2.10.2 Nonoperating conditions. Nonoperating conditions shall be as specified in a and b:

a. Ambient temperature: -62°C to +71°C
b. Elevation: Sea level to 50,000 ft

3.2.10.3 Dust. The RCS shall be constructed to withstand the dust test specified in 4.5.2 without performance degradation or malfunction.

3.2.11 Test points. The RCS shall be provided with adequate and well-defined test points to facilitate making performance measurements and for use in maintaining the RCS.

3.2.11.1 Circuit test points. Circuit test point accesses shall be provided which comply with the criteria specified in MIL-STD-415. Test points shall be provided for the tests specified in a through g:

a. Signal quality
b. Circuit performance
c. System alignment
d. Installation and checkout
e. Planned maintenance systems test
f. Line replaceable unit (LRU) fault location
g. Shop replaceable part (SRP) fault isolation

Test points shall be provided with proper interface fittings and matching circuits to connect the selected support test equipment.

3.2.11.2 Support test instrumentation. Support test instrumentation shall be as specified in 3.2.11.2.1 and 3.2.11.2.2.

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3.2.11.2.1 Line operation and maintenance test support. The selection of general purpose electronic test equipment (GPETE) shall be determined to perform the tests specified in a through d:

a. Signal quality
b. Circuit performance
c. Adjustments
d. Calibration and LRU fault location

3.2.11.2.2 GPETE. GPETE shall be selected from MIL-STD-1364 and TABLE I.

TABLE I. GPETE.

Test equipment	Description
AN/GGM-15	Distortion analyzer and toggle signal generator (same as CCZB-DT9600C)
AN/USM-181A	Telephone test set
AN/USM-207	Frequency counter
AN/USM-296	Oscilloscope
AN/USM-329	Distortion analyzer and toggle signal generator
CAQI-140B	Oscilloscope-spectrum analyzer
CAQI-651B	Test oscillator
CAQI-7000A/17005A	X-Y recorder with paper take-up roller
CAQI-7560A	Logarithmic converter for X-Y recorder, dual channel
CAQI-E03-8553B	Spectrum analyzer assembly
CAQI-8443A	Tracking signal generator counter
CDDD-TTS-58B	Noise measuring set and impulse noise counter
CDDD-TTS-85CR	Transmission measuring set
CDJQ-44	Phase jitter meter, portable
CDJQ-45	Phase jitter meter, rack-mounted
CDSD-PSA-031	Plug-in, spectrum analyzer, low frequency (for use with CAQI-140B main frame)
CN-947/USM	Impedance matching transformer
SG-860/GGM(V)	Data and telegraph message generator
SG-885(P)/USM	Digital distortion and message generator
TS-1467A/URM-114	Radio calibrator set
TS-1830/U	Audio wave analyzer
TS-2395/G	Envelope delay test set
TS-2778/USM	Audio level measuring set
TS-2908(P)/UGM	Telegraph data analyzer, carrier
AN/UIC-5	Secure intercom system
AN/URQ-10A	Frequency standard
AN-FSQ-44	Clock, master, for 9.6 kbps 1/ modem, with stability of ±1 part in 10^8 per day

1/ Kilobits per second

3.2.12 Ventilation and cooling. The RCS shall be ventilated and cooled in such a manner that, under normal conditions of operation at the maximum ambient temperature of +50°C, no part shall attain a temperature which will tend to cause damage or reduce the useful life of the RCS.

3.2.13 Test cards and module extenders. All test cards and printed cards or module extenders required for operation and maintenance of components of the RCS shall be provided with the RCS. Each module shall be completely removable by single disconnect. Signal and power for servicing and testing of modules in the extended position shall be provided by the basic unit under test. The power supply shall be exempted from this requirement.

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3.2.14 Thermal survey. The contractor shall conduct a complete thermal survey of the first article sample RCS to determine the existence of hot spots which may prove to be injurious to the useful life of the RCS. The complete thermal survey shall be conducted with the equipment cabinets closed. The survey shall be completed and all required corrective measures taken by the contractor prior to acceptance of the first article sample by the Government.

3.2.14.1 Definition. For the purpose of this specification, a hot spot is any component or part used in the RCS including, but not limited to, such devices as transistors, light emitting diodes, bulbs, fuses, diodes, integrated circuits, printed circuit boards, insulation, insulators, transformers, capacitors, chokes, coils, and resistors which attain a temperature in excess of that recommended by the component or part manufacturer, or by any applicable military specification, or that attains a temperature which causes accelerated deterioration.

3.2.14.2 Survey conditions. The thermal survey shall be made while the RCS is continuously operating in an environmental temperature of 50°C and shall begin immediately after the RCS has thermally stabilized for a minimum of 24 hours at that temperature. The contractor shall take the necessary measures to prevent the environmental temperatures in which the thermal survey is being conducted from decreasing to less than 48°C while placing or rearranging the thermal sensing device(s) from one component or part to another. The RCS shall be fully operational during the entire thermal survey, if practical. If it should become necessary to remove the alternating current (AC) power from the RCS for more than one period of up to 5 minutes per hour, the RCS shall be thermally stabilized for at least 4 hours before proceeding with the survey. The complete thermal survey shall be conducted with the AC input voltage set at 10 percent above the nominal specified voltage for the RCS under test. The AC line frequency shall be set for 47 hertz (Hz). The contractor shall accurately measure and record all thermal data pertaining to each component or part specified herein unless specifically exempted, in writing, from such measurement by the procuring activity. The thermal data shall include, but not be limited to, the actual operating temperature of the part or component and any visible or electrical evidence, or both, of deterioration of the RCS.

3.2.14.3 Comparison. The contractor shall provide a comparison between the thermal data taken during the survey and the thermal conditions recommended by the manufacturer of the part or component. The method used to measure the operating temperature of the component shall not affect the accuracy of the measurement as a result of the heat sink effect, or for any other reason.

3.2.14.4 Proper techniques. The contractor shall use proper techniques such as thermocouples, infrared photography, chemicals, or calibrated thermally sensitive materials, to accurately measure the temperature of each component or part included in the survey.

3.2.14.5 Waiver of survey. When it is determined by proper engineering analysis that a particular component or part could be exempted from the survey, the contractor may request a waiver from the procuring activity which, if granted, will exempt those items from the survey. In the request for waiver, the contractor shall provide a complete engineering rationale for excluding the specific component(s) from the thermal survey.

3.2.15 Mounting. Each unit of the RCS shall be configured for mounting in a standard 19-inch (in.) rack in accordance with EIA-RS-310-C-77. The mounting shall provide ready access to the interior of each unit for maintenance and servicing. The front panels of the local control multiplexer and the local control interface unit shall be arranged in such a manner that when the panels are opened all their printed circuit cards (PCC) and modules are exposed for servicing. When viewed from the front, the panel hinges shall be located on the right side of the units. The RCU and the display and entry units (DEUs) shall be mounted on slides to permit withdrawal of the units for servicing. Withdrawal of these units shall provide total access to the units and their PCCs.

3.2.16 Front panel arrangement. Front panel controls, indicator lights, switches, displays, and markings, shall be in accordance with MIL-STD-454, Requirements 28, 50, 58, and 67. Panel layouts which are identical to any GFE AN/FSQ-98 (see 6.7) provided on this contract are acceptable to the Government.

3.2.17 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of the item or the equipment. Subassemblies shall be interchangeable with those in the GFE, if provided.

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3.2.18 Mode test switch. The RCU and the local control multiplexer unit shall each be provided with an internal two-position mode test switch that will permit two modes of operation for these units. With these test switches it shall be possible to operate and test any external data loop used in the RCS as specified herein. The switch positions shall be labeled 1 and NORMAL. When the switch is placed in the 1 position, the unit shall be forced to a sustained logical 1 condition. When the switch is placed in the NORMAL position the forced logical 1 condition shall be removed and the RCS shall function normally. In order to ensure that equipment operating tolerances are maintained at all times, the output data loop currents produced during a forced logical 1 condition shall be equal to loop currents produced during NORMAL operation. This requirement applies to both the low level polar and the neutral modes of operation. The purpose of this feature of the equipment is to permit accurate and speedy adjustment of neutral loop currents and monitoring of polar loop voltage. The switch shall be accessible in each equipment without removal of any component in the unit.

3.2.19 Line filters. The basic components comprising the complete RCS shall contain line filters sufficient to conform to the performance requirements specified in 3.4.1.

3.3 RCS definition. The RCS shall be capable of performing the functions specified in a through d:

a. Providing for the remote serial control of up to 10 AN/FRT-84 transmitters from a remotely positioned control unit (see 3.4.1.1.)
b. Selecting any frequency or mode available on AN/FRT-83, AN/FRT-84, and AN/FRT-85, radio transmitters
c. Providing read-back information from a selected transmitter to the remotely positioned control unit to display the status of remotely or automatically controlled transmitter functions as specified
d. Providing continuous fault monitoring of all controlled transmitters and the associated data channels

3.3.1 Major components. The RCS shall consist of the major components specified in a through c:

a. Remote control group consisting of one Control Unit, Remote, C-9427/FSQ-98(V) and one DEU ID-1925/FSQ-98(V). (The RCU can be equipped with up to 10 DEUs)
b. Interface unit - Local control, J-3200/FSQ-98(V)
c. Multiplexer - Demultiplexer, TD-1102/FSQ-98(V)

3.3.2 RCS interface definition. The RCS shall transmit and receive all remote control messages and read-back messages on a single full duplex data circuit (two pairs of wires) between the RCU and the transmitter(s) end of the system.

3.4 Characteristics. RCS characteristics shall be as specified in 3.4.1 through 3.4.3.4.

3.4.1 Performance. The RCS shall conform to the performance requirements specified in 3.4.1.1 through 3.4.1.11.6.

3.4.1.1 Serial control system. The RCS shall be capable of exercising serial control of up to 10 AN/FRT-84 transmitters through one RCU and up to 10 DEUs, one for each transmitter. At the transmitter station the local control multiplexer shall interface with up to 10 local control interface units which shall be mounted in the individually controlled transmitters. Each system shall be capable, without field modification, of accommodating up to eight local control multiplexers. Systems operation shall be synchronous with communications discipline maintained by the RCU.

3.4.1.1.1 Serial message categories. The serial control system shall be implemented by three categories of messages (or words) sent or received by the RCU (see TABLE II) as specified in a through c:

a. Control messages. The RCU shall generate control messages either as a result of inputs from the DEUs in conjunction with its operator request capability or as a result of preprogrammed routines internal to the RCU.
b. Monitor request message. The RCU shall request system information from the local control multiplexer to supply operational condition data to the DEU.
c. Monitor reply message. The local control multiplexer shall transmit the status of the transmitter or other system aspect. The local control multiplexer can originate a monitor reply message.

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TABLE II. Message word format.

ASCII 1/ Format - data link control request word									
First character	SOM 2/								
	Symbol	b8	B7	B6	B5	B4	B3	B2	B1
SOM (SOH) 3/		0	0	0	0	0	0	0	1
Second character	Address								
	Symbol	B8	B7	B6	B5	B4	B3	B2	B1
0	0	1	0	1	1	0	0	0	0
1	1	0	0	1	1	0	0	0	1
2	2	0	0	1	1	0	0	1	0
3	3	1	0	1	1	0	0	1	1
4	4	0	0	1	1	0	1	0	0
5	5	1	0	1	1	0	1	0	1
6	6	1	0	1	1	0	1	1	0
7	7	0	0	1	1	0	1	1	1
8	8	0	0	1	1	1	0	0	0
9	9	1	0	1	1	1	0	0	1
Third character	System instruction								
	Symbol	B8	B7	B6	B5	B4	B3	B2	B1
Command	B	1	1	0	0	0	0	1	0
monitor	A	0	1	0	0	0	0	0	1
status	C	0	1	0	0	0	0	1	1
Fourth character	Sub address								
	Symbol	B8	B7	B6	B5	B4	B3	B2	B1
0	0	1	0	1	1	0	0	0	0
1	1	0	0	1	1	0	0	0	1
2	2	0	0	1	1	0	0	1	0
3	3	1	0	1	1	0	0	1	1
4	4	0	0	1	1	0	1	0	0
5	5	1	0	1	1	0	1	0	1
6	6	1	0	1	1	0	1	1	0
7	7	0	0	1	1	0	1	1	1
8	8	0	0	1	1	1	0	0	0
9	9	1	0	1	1	1	0	0	1
Fifth character	Decoding instructions								
	Symbol	B8	B7	B6	B5	B4	B3	B2	B1
Tune command	A	1	1	0	0	0	0	0	1
Change frequency	E	0	1	0	0	0	1	0	1
Change emission	K	1	1	0	0	1	0	1	1
Change mode	B	1	1	0	0	0	0	1	0
Sixth character	Class of emission								
	Symbol	B8	B7	B6	B5	B4	B3	B2	B1
A0	0	1	0	1	1	0	0	0	0
A2A3H	2	0	0	1	1	0	0	1	0
A1F1F4	4	0	0	1	1	0	1	0	0
SSB-10	6	1	0	1	1	0	1	1	0
SSB-20	8	0	0	1	1	1	0	0	0
SSB-40	:	1	0	1	1	1	0	1	0
SSB-00	>	0	0	1	1	1	1	1	0

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TABLE II. Message word format. (continued)

ASCII [1/] Format - data link control request word

Seventh character — Mode

	Symbol	B8	B7	B6	B5	B4	B3	B2	B1
Operate	1	0	0	1	1	0	0	0	1
Operate + USB [4/]	3	1	0	1	1	0	0	1	1
Operate + USB or PTT [5/] or both	5	1	0	1	1	0	1	0	1
Operate + USB or VFG [6/] or both	7	0	0	1	1	0	1	1	1
Operate + LSB [7/]	9	1	0	1	1	1	0	0	1
Operate + ISB (2)	;	0	0	1	1	1	0	1	1
Operate + ISB (4)	=	0	0	1	1	1	1	0	1
Standby	Ø	1	0	1	1	0	0	0	0
Standby + USB	2	0	0	1	1	0	0	1	0
Standby + USB or PTT, or both	4	0	0	1	1	0	1	0	0
Standby + USB or VFG, or both	6	1	0	1	1	0	1	1	0
Standby + LSB	8	0	0	1	1	1	0	0	0
Standby + ISB (2)	:	1	0	1	1	1	0	1	0
Standby + ISB (4)	<	1	0	1	1	1	1	0	0

Eighth character — 10 MHz digit

	Symbol	B8	B7	B6	B5	B4	B3	B2	B1
0	0	1	0	1	1	0	0	0	0
1	1	0	0	1	1	0	0	0	1
2	2	0	0	1	1	0	0	1	0

Ninth character — 1 MHz digit

0	0	1	0	1	1	0	0	0	0
1	1	0	0	1	1	0	0	0	1
2	2	0	0	1	1	0	0	1	0
3	3	1	0	1	1	0	0	1	1
4	4	0	0	1	1	0	1	0	0
5	5	1	0	1	1	0	1	0	1
6	6	1	0	1	1	0	1	1	0
7	7	0	0	1	1	0	1	1	1
8	8	0	0	1	1	1	0	0	0
9	9	1	0	1	1	1	0	0	1

Tenth character — 100 kHz digit

Eleventh character — 10 kHz digit

Twelfth character — 1 kHz digit

Same form as ninth character

Thirteenth character — 0.1 kHz digit

Monitor request word

First character — SOM

Same as first character of control request word

Second character — Address

Same as second character of control request word

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TABLE II. Message word format. (continued)

Third character	System instructions								
	Symbol	B8	B7	B6	B5	B4	B3	B2	B1
Monitor	A	0	1	0	0	0	0	0	1
Fourth character	Sub address								
Same as fourth character of control request word									
ASCII format - monitor reply word									
First character	SOM								
Same as first character of control request word									
Second character	Station address								
Same as second character of control request word									
Third character	Transmitter sub address 0 status								
	Symbol	B8	B7	B6	B5	B4	B3	B2	B1
Operate	3	1	0	1	1	0	0	1	1
System tuning	1	0	0	1	1	0	0	0	1
Standby or ready	6	1	0	1	1	0	1	1	0
Standby	4	0	0	1	1	0	1	0	0
Fault	7	0	0	1	1	0	1	1	1
Local control	5	1	0	1	1	0	1	0	1
Fourth character	Transmitter sub address 1 status								
Fifth character	Transmitter sub address 2 status								
Sixth character	Transmitter sub address 3 status								
Seventh character	Transmitter sub address 4 status								Same form
Eighth character	Transmitter sub address 5 status								as third
Ninth character	Transmitter sub address 6 status								character
Tenth character	Transmitter sub address 7 status								
Eleventh character	Transmitter sub address 8 status								
Twelfth character	Transmitter sub address 9 status								

1/ American Standard Code For Information Interchange
2/ Start of message
3/ Start of heading
4/ Upper sideband
5/ Push to talk
6/ Voice frequency generator
7/ Lower sideband

3.4.1.1.2 Control signal electrical characteristics. Control signal electrical characteristics shall be as specified in a through d:

a. Rate. Signalling rates between the RCU and the local control multiplexer shall be selectable by switches located in the RCU and in the local control unit. These switches shall provide selection of 75 baud, 150 baud, 300 baud, and 600 baud rates. If both the RCU and the local control multiplexer are not set to provide the same signalling rate, a system data fault shall be indicated and no control commands shall be passed to the controlled transmitters.

b. Encoding. Binary code, nonreturn to zero, equal unit intervals coding shall be used for signalling throughout the RCU.

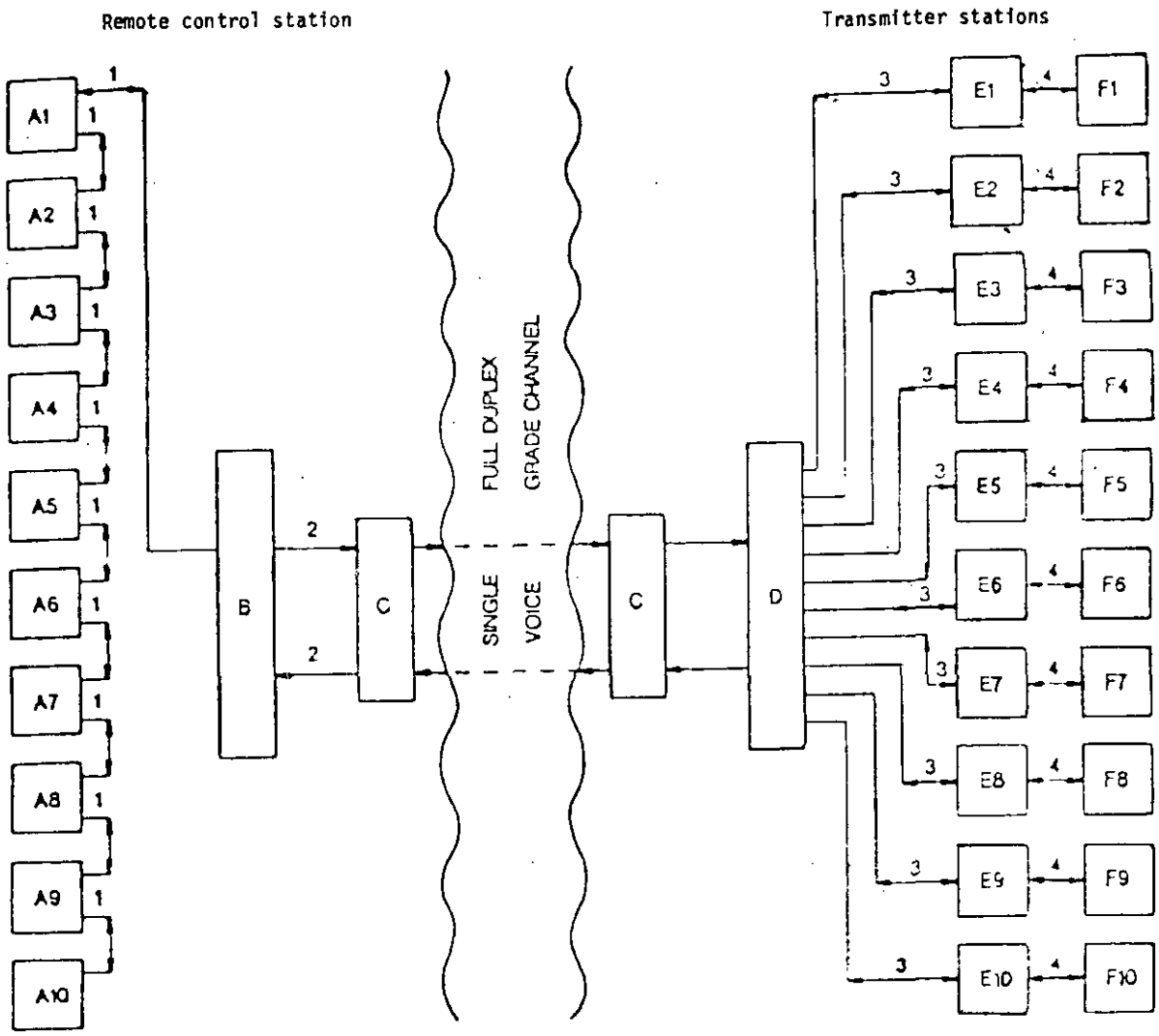
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c. Levels. The RCS shall include provisions for both low-level polar and high-level neutral signalling loops between the RCU and the local control multiplexer as shown in FIGURE 1. Where FIGURE 1 indicates high-level neutral or low-level polar signalling, the appropriate units shall include switches so that either signalling level may be selected. These switches shall be located inside the units and readily accessible when the units are opened. For purposes of this specification, high-level neutral signalling is 60 milliamperes (mA) of direct current (DC) during a logical 1 interval and 0 mA during logical 0 interval. Each high-level neutral receive function in the RCS shall respond to an external loop signal of 60 mA ±10 mA for logical 1 condition. Low-level polar signalling in this specification is +6 volts direct current (VDC) ±1 VDC during a logical 1 interval and -6 VDC ±1 VDC during a logical 0 interval. For low-level operation, loop current in any loop shall not exceed 1 mA during normal operation of the equipment. Signalling between the DEU and RCU, and between the local control multiplexer and the local control interface units shall use polar signalling +4 VDC ±1 VDC defining a logical 1 state and -4 VDC ±1 VDC defining a logical 0 state. This equipment shall be compatible with the requirements specified in the DC Digital Transmission Reference Systems paragraph of MIL-STD-188.

d. Signal power supply. Each system signal source shall include its own isolated 130 VDC power supply to supply the external loop circuit (in the case of high-level operation). Each system signal source shall also include an appropriate low-level power supply for low-level loop operation. Each high-level system signal source shall include provision for adjustment of loop current to the specified value. The loop current adjustment control shall provide capability to adjust loop current to 60 mA when the external loop resistance is any value between 0 ohms and 1500 ohms.

3.4.1.1.3 Input/output (I/O) characteristics. Interfaces between the units of the RCS and the attendant transmitter shall be in accordance with the connector and terminal designations of TABLES III through VII. Block C of FIGURE 1 represents standard Navy voice-frequency-carrier-telegraph (VFCT) units not included in this specification nor provided as GFE. Blocks B and D of FIGURE 1 are basically data processing devices; all transmitter control and transmitter status indications are via the DEU (block A). The remote control station in FIGURE 1 utilizes a maximum of 10 DEUs to control 10 transmitters collocated in the same installation site. Connectors are provided on the DEU specifically to connect the units in parallel, with one unit connected directly to the RCU. The signal lines which connect the RCU and the local control multiplexer units to the VFCT unit shall be balanced and ungrounded. FIGURE 2 illustrates a second configuration using the RCS with transmitters located at separate stations. As specified in 3.4.1.1, each system is capable of accommodating up to eight local control multiplexers and, in this example, 10 transmitters located at eight separate stations are controlled from a single remote site. This configuration requires a local control multiplexer and a VFCT unit at each transmitter site. Any other combination of local control multiplexers and local control interface units to control up to 10 transmitters is permissible, for example, two stations with one multiplexer and five interface units (five transmitters per station) at each station.

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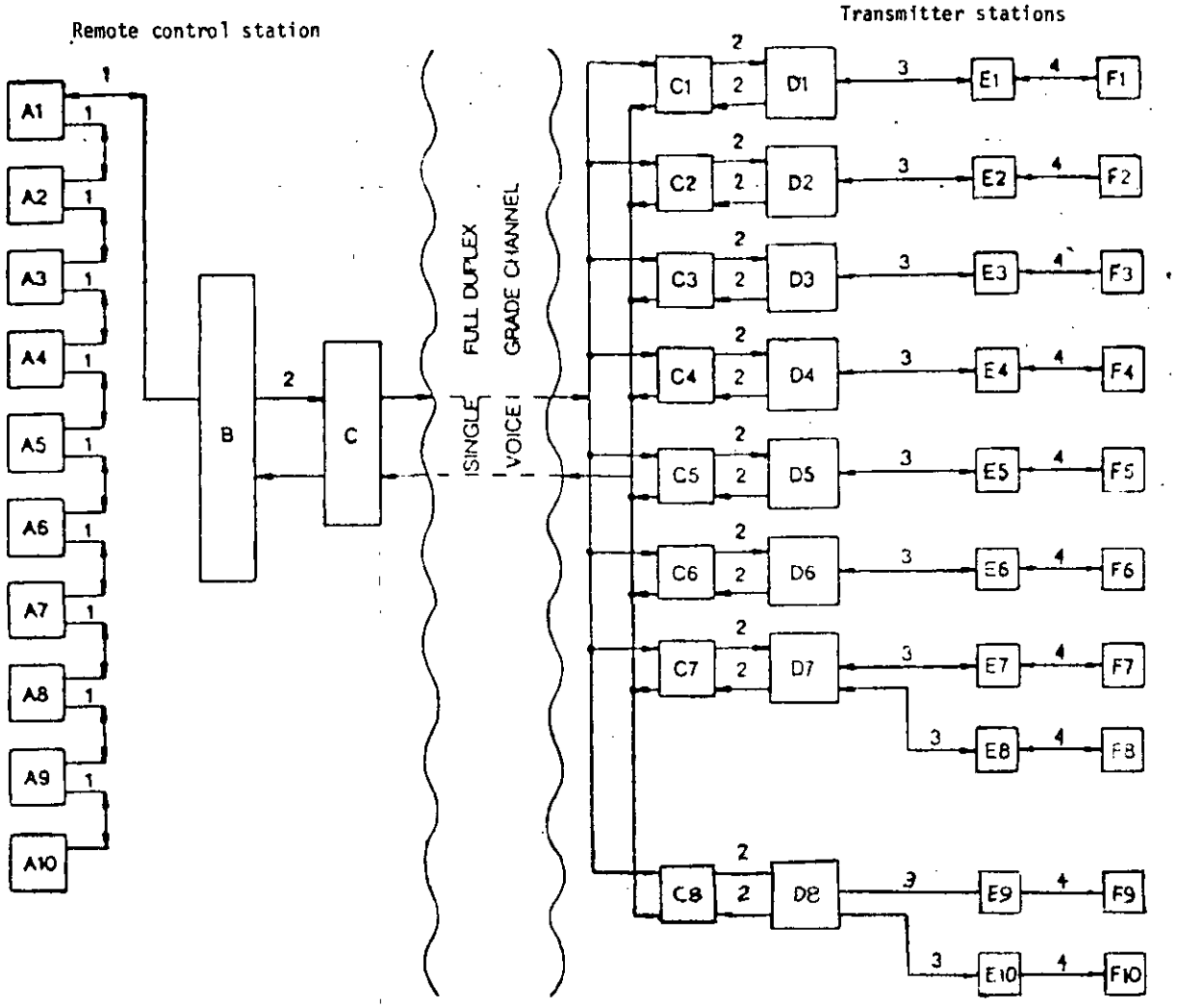


A DEU
B RCU
C VFCT equipment
D Multiplexer - demultiplexer
E Interface unit - local control
F Controlled transmitters (with DEU each)

1 - Multiwire cable, DEU to remote control and control loop
2 - Single pair of wires, high-level neutral or low-level polar signals
3 - Multiwire cable, multiplexer to interface unit control, 300 ft maximum cable length
4 - Multiwire cable interface unit to controlled transmitter

FIGURE 1. Typical remote control station (transmitters at single installation site).

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A DEU
B RCU
C VFCT equipment
D Multiplexer - demultiplexer
E Interface unit - local control
F Controlled transmitters (with DEU each)

1 - Multiwire cable, DEU to remote control and control loop
2 - Single pair of wires, high-level neutral or low-level polar signals
3 - Multiwire cable, multiplexer to interface unit control, 300 ft maximum cable length
4 - Multiwire cable interface unit to controlled transmitter

FIGURE 2. Typical remote control station (multiple site transmitter installation).

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TABLE III. DEU panel connectors.

Function	Connector type	Terminal assignment	Notes
AC power input	MS3102A-10SL-3P	A 115 VAC 1/ B Ground C 115 VAC common	
Signal interface to RCU	M24308/4-3		
Signal interface to additional DEU	M24308/2-3		

1/ Volts alternating current

TABLE IV. RCU connectors.

Function	Connector type	Terminal assignment	Notes
AC power input	MS3102A-14S-7P	A 115 VAC B Ground C 115 VAC common	
Signal interface to DEU	M24308/2-3		
Command signal output	MS3102A-14S-7S	A Ground B TTY 1/ common C TTY high	
Status signal input	MS3102A-14S-7PW	A Ground B TTY common C TTY high	

1/ Teletype

TABLE V. Local control multiplexer connectors.

Function	Connector type	Terminal assignment	Notes
AC power input	MS3102A-14S-7P	A 115 VAC B Ground C 115 VAC common	
Command signal input	MS3102A-14S-7PW	A Ground B TTY common C TTY high	
Status signal output	MS3102A-14S-7S	A Ground B TTY common C TTY high	
Signal interface to RCUs (10 connectors)	MS3112P-14-15S		

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TABLE VI. Local control interface unit connectors.

Function	Connector type	Terminal assignment	Notes
AC power input	MS3102A-14S-7P	A 115 VAC B Ground C 115 VAC common	
Signal interface to multiplexer	MS3102A-20-27P		
Signal interface to MD-777/FRT J-7	MS3114E-20-39P	A B C 100 Hz D E F G 1 kHz H J K L 10 kHz M N P R 100 kHz S T U V 1 MHz W X Y 10 MHz Z VCO 1/ A a VCO B b AØ c A2, A3H d A1, F1, F4 e SSB 2/ -10 dB 3/ carrier f SSB -20 dB carrier g SSB -40 dB carrier h SSB -00 carrier i Standby command j Operate command k Ground m Ground n PTT command	4/
Signal interface to MD-777/FRT J-8	MS3114E-20-39PW	A USB select control B LSB select control C ISB (2) select control D ISB (4) select control G AØ class of emission read-back H A_2, A3H class of emission read-back J A_1, F_1, F_4 class of emission read back K SSB -10 class of emission read-back L SSB -20 class of emission read-back M SSB -40 class of emission read-back N SSB - class of emission read-back	5/

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TABLE VI. Local control interface unit connectors. (continued)

Function	Connector type	Terminal assignment	Notes
		P Operate read-back R Standby read-back S USB read-back T LSB read-back U ISB (2) read-back V ISB (4) read-back Y Ready read-back (common) Z Ready read-back a Fault read-back b Fault read-back (common) c Local status d Transmitter not available read-back (common) e Tune command f Tune activate common (ground) g +5 VDC h PTT read-back i Transmitter fail j VFG read-back k Standby and operate read-back (common) m sideband select (common) n Sideband read-back (common) q Class of emission read-back (common) v VFG command	5/

1/ Voltage controlled oscillator
2/ Single sideband
3/ Decibel
4/ For control code see TABLE VII
5/ For control code see GFE

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TABLE VII. Unit connector frequency selection.

100 Hz digit selection 1/					1 kHz digit selection				
Digit	J7-A	J7-B	J7-C	J7-D	Digit	J7-E	J7-F	J7-G	J7-H
0	0	0	0	1	0	1	0	0	1
1	1	0	0	1	1	0	0	0	1
2	0	0	1	1	2	1	0	1	1
3	1	0	1	1	3	0	0	1	1
4	0	1	1	0	4	1	1	1	0
5	1	1	1	0	5	0	1	1	0
6	0	1	0	0	6	1	1	0	0
7	1	1	0	0	7	0	1	0	0
8	0	0	0	0	8	1	0	0	0
9	1	0	0	0	9	0	0	0	0
10 kHz digit selection					100 kHz digit selection				
Digit	J7-J	J7-K	J7-L	J7-M	Digit	J7-N	J7-P	J7-R	J7-S
0	0	0	1	1	0	0	0	0	1
1	1	1	0	1	1	1	0	0	1
2	0	1	0	1	2	0	0	1	1
3	1	0	0	1	3	1	0	1	1
4	0	0	0	1	4	0	1	1	0
5	1	1	1	0	5	1	1	1	0
6	0	1	1	0	6	0	1	0	0
7	1	0	1	0	7	1	1	0	0
8	0	0	1	0	8	0	0	0	0
9	1	1	0	0	9	1	0	0	0
1 MHz digit selection					10 MHz digit selection				
Digit	J7-T	J7-U	J7-V	J7-W	Digit	J7-X	J7-Y		
0	1	1	0	0	0	1	1		
1	0	1	0	0	1	0	1		
2	1	0	0	0	2	1	0		
3	0	0	0	0					
4	1	0	0	1					
5	0	0	0	1					
6	1	0	1	1					
7	0	0	1	1					
8	1	1	1	0					
9	0	1	1	0					

VCO 1/ selection		
Frequency range (MHz)	J7-Z (VCO A)	J7-a (VCO B)
2.0000 to 11.9999	0	0
12.0000 to 21.9999	0	1
22.0000 to 29.9999	1	0

1/ The information contained in this TABLE defines the binary stimulus required by the MD-777 exciter for the desired digit selection.

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3.4.1.1.4 Message word format. The format for the control, monitor request, and monitor reply words shall be in accordance with TABLE II.

3.4.1.2 Data link channel. The data link channel shall consist of one full duplex circuit for control and monitoring of the AN/FRT-84 transmitter(s) when terminal equipment is utilized for microwave or telephone cable transmission of control data.

3.4.1.2.1 Transmission of control and read-back data. The necessary data to control each of the associated transmitters shall be transmitted over a single pair of wires. All read-back data from the multiplexer unit to the RCU shall be transmitted over a single pair of wires.

3.4.1.2.2 System distortion. The RCS shall be designed to operate properly with a high degree of distortion present on any loop used in the control system. When the RCS is connected in a back-to-back configuration, the RCS shall perform normally with a distortion level of 0 percent to 35 percent on any control loop.

3.4.1.3 Data link error control and monitoring. The methods of error detection and monitoring specified in 3.4.1.3.1 through 3.4.1.3.5.1 shall be implemented.

3.4.1.3.1 Parity. The last bit of each character shall be encoded to achieve an odd parity for the preceding seven bits of the character. If even parity of any character occurs within a control message, the message contents shall not be processed.

3.4.1.3.2 Address comparison. The RCU shall compare the address character of the local control unit response for a one-to-one correspondence to the originally transmitted address. An invalid comparison shall prevent message processing.

3.4.1.3.3 Message comparison. Each control command word transmitted by the RCU shall be echoed by the addressed local control unit back to the RCU. The RCU shall compare the returned data for a one-to-one correspondence to the originally entered and transmitted control data. Upon receipt of an error-free comparison, the RCS shall revert to the normal monitor mode. If the controlled transmitter is still tuning when the RCS reverts to the monitor mode, a tuning indicator on the DEU associated with the transmitter shall be illuminated.

3.4.1.3.4 Channel monitoring. The RCS shall perform the data channel failure monitoring specified in a and b.

a. No response. The local control multiplexer addressed by the RCU shall respond within 30 milliseconds (ms) after receipt of the termination of the RCU transmission. After termination of the transmission cycle, the RCU shall monitor for the start of the multiplexer response. This RCU monitor time frame shall be of fixed duration, once adjusted for the data channel delay as specified in 3.4.1.6. If the addressed multiplexer does not respond within the predetermined time frame, the automatic request (ARQ) circuits of the RCU shall be activated to transmit an identical message to the nonresponsive multiplexer. A second no-response shall activate the RCU data channel error monitor circuitry.

b. No request. Each transmission from the RCU shall be decoded by each multiplexer regardless of the message address, and each multiplexer shall monitor the time frame occurring between RCU data transmissions. Exceeding this time frame shall be recognized by the multiplexer as a data channel loss. Time frame monitoring shall be adjustable to conform to the propagation delay times specified in 3.4.1.6.

3.4.1.3.5 ARQ. Detection of an error by the RCU in the data transmission, whether the error is due to parity, address comparison, message comparison, improper SOM character detection, or a no-response condition, shall cause the RCU to transmit a new message in accordance with 3.4.1.3.5.1. If the second message cycle is not error-free, the RCU shall activate the appropriate error monitor, dependent upon the system mode of operation.

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3.4.1.3.5.1 Control word encoding. The data instruction character of a control word shall be encoded automatically as a result of operator selection of control parameter settings. The local control interface unit shall decode this character to determine which parameters in a control word message should be processed. In this manner if an error occurs in an unchanged control parameter, the error shall be ignored as processing of this parameter has automatically been inhibited. This process shall be invoked during the ARQ mode of operation. The echoed message received from the local control interface via the multiplexer unit shall undergo one-to-one comparison with operator-selected control parameter information. Only parameters which are not in agreement during this comparison shall be processed on the second transmission of the control data as a result of the now modified coding of the data instruction character.

3.4.1.4 Synchronization. Synchronization for proper bit decoding shall be recovered from the incoming data by both the RCU and local control unit. Synchronization of the multiplexer unit shall be accomplished by incoming data from the RCU in a manner compatible with system requirements.

3.4.1.5 Language. Each character of a control or monitor word shall be coded to conform to the eight-level ASCII code as specified in MIL-STD-188.

3.4.1.6 Propagation delays. The RCS shall be designed to permit satisfactory operation in communication links which employ long radio frequency (RF) paths (satellite) and VFCT equipment. The absolute propagation delay associated with a particular data link in which this VFCT equipment is to be used will vary as a function of the number of VFCT units employed in the entire system and the length of the free-space RF paths. The total time allowed for the round trip propagation time delays shall be divided into five selectable increments of 0 second to 0.428 second, 0 second to 1.71 seconds, 0 second to 3.42 seconds, and 0 second to 6.84 seconds. The RCU shall be designed to properly process read-back messages from the local control multilplexer that have been delayed by any time interval that is greater than 0 seconds and less than the maximum time interval selected for a particular data link. These time increments, which compensate for long propagation delay, shall be selectable by appropriate switches located internal to the RCS.

3.4.1.6.1 Polling time requirements. Notification of the failure of the communications channel between the RCU and any local control multiplexer shall occur within a time period equal to the sum of the time duration of the command word, and two times the preset propagation delay. When the RCS includes one multiplexer unit, the failure of any controlled transmitter shall be visually displayed at the DEU, and an audible alarm activated within a time period equal to the sum of the time duration of the monitor request word, the monitor reply word, the response time of the multiplexer, and the preset propagation delay time. The multiplexer response time shall be as specified in 3.4.1.3.4a, the time duration of the monitor and command words shall be the reciprocal of the signalling rate, the bit length of the word (see TABLE II), and the propagation delay shall be as specified in 3.4.1.6. Additional multiplexers in the RCS shall increase the total transmitter fault polling time by not more than the number of multiplexers times the allowable time duration for one multiplexer.

3.4.1.7 Primary power. All units of the RCS shall operate independently from a 115 VAC or 230 VAC ±10 percent, 47 Hz to 63 Hz, single-phase power source.

3.4.1.7.1 Power switch. Each unit of the RCS other than the DEU shall be provided with a front panel toggle switch to control the application of primary power.

3.4.1.7.2 Primary power protection. Each unit of the RCS other than the DEU shall be provided with front panel-mounted fuseholders and appropriately rated fuses for primary power protection.

3.4.1.7.3 Power supplies. All power supplies for the logic circuitry of the RCS shall contain overcurrent protection. Each of these power supplies shall be capable of operating into a short-circuit for 5 minutes at an equipment ambient temperature of 50°C without failure.

3.4.1.7.4 Loss of power. Loss of power to the local control multiplexer or to the local control interface unit shall necessitate the reprogramming of the system when it resumes operation.

3.4.1.8 DEU. The DEU shall be used to control and monitor the frequency and mode of operation of one AN/FRT-83(V), AN/FRT-84(V), and AN/FRT-85(V) transmitter from a remote location. The DEU shall connect directly to the RCU to form the remote control group.

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3.4.1.8.1 Control functions. The DEU shall provide the data entry and system control switches specified in a through e to perform the functions specified for the controlled transmitters.

a. Frequency select. The frequency select switch shall be a six-decade thumbwheel switch used to select any of the 280,000 transmitter operating frequencies from 2.0000 MHz to 29.9999 MHz in 100 Hz increments. This frequency selection shall be labeled with digits corresponding to the selected frequency in MHz.

b. Sideband select. The sideband select switch shall be a six-position thumbwheel switch for selecting the sideband modes specified in 1 through 6:

1. USB
2. USB PTT
3. USB VFG
4. LSB
5. ISB 2
6. ISB 4

c. Class of emission select. The class of emission select switch shall be a seven-position thumbwheel switch for selecting the classes of emission specified in 1 through 7:

1. Ao
2. A_2/A_3E
3. $A_1/F_1/F_4$
4. SSB/-10 dB
5. SSB/-20 dB
6. SSB/-40 dB
7. SSB/-

When the Ao, $A_1/F_1/F_4$, or A_2/A_3E classes of emission are selected, the USB mode shall have priority over the sideband selection.

d. Standby/operate switch. The standby/operate switch shall be a two-position toggle switch for selection of either a standby or an operate condition for the addressed transmitter.

e. Activate switch. An illuminated pushbutton switch shall be provided for controlling information being programmed to the selected transmitter. It shall be possible from the remote location to program the RCS for any frequency or mode of operation available on the AN/FRT-84 radio transmitters without the transmitters actually changing frequency or mode of operation until the activate switch has been depressed. The switch shall be internally illuminated with a white light during the control command cycling period of the control system. The command cycling period is the time required for the command word to be sent to the controlled transmitter, the new command parameters read back to the DEU and a one-for-one comparison of data made. The command cycle shall end when the data comparison is error-free. The switch shall be spring-loaded to return to its neutral position when the energizing force is removed. The switch shall be labeled ACTIVATE.

3.4.1.8.2 Status indicators. The DEUs shall provide the indicators specified in a through g to display the status of the controlled transmitters:

a. Ready or local. The ready or local indicator shall be a split screen indicator; the upper portion shall be internally illuminated with a green light and shall indicate the ready status of the addressed transmitter; the bottom portion shall be internally illuminated with a blue light and shall indicate whether the addressed transmitter is in local control. The green portion, labeled READY, shall light to indicate completion of the addressed transmitter tuning to the frequency indicated on the DEU. The blue portion, labeled LOCAL, shall light to indicate that the addressed transmitter has been switched to local control and cannot be controlled by inputs from the DEU. The DEU shall continue to indicate the status of the addressed transmitter (except frequency), even though the transmitter is being locally controlled.

b. Fault and tuning. The fault and tuning indicator shall be a split screen indicator: The upper portion shall be internally illuminated with a red light and shall indicate that the transmitter is in a tuning process. The red portion, labeled FAULT, shall light to indicate an operational fault of any nature in the associated transmitter. The lower yellow portion, labeled TUNING, shall light only while the associated transmitter is in a tuning process.

c. Standby and operate. The standby and operate indicator shall be a split screen indicator: The upper portion shall be illuminated with a green light; the lower portion shall be illuminated with a yellow light. The green portion, labeled OPERATE, shall light to indicate that the addressed transmitter is in an operate condition. The yellow portion, labeled STANDBY, shall light to indicate that the addressed transmitter is in a standby condition. When the address switch is changed to a different address, the standby and operate indicator shall indicate the conditions of the newly addressed transmitter without the necessity of depressing the status switch.

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d. Valid emission indicator. The valid emission indicator shall be located adjacent to the emission select switch to denote that the class of emission shown on the switch has been properly processed at the transmitter site. This indicator shall be labeled VALID, and shall be illuminated by a yellow lamp.

e. Fixed mode indicator. A fixed mode indicator shall be provided adjacent to the mode select switch and shall be illuminated with a yellow lamp. Turning this indicator on indicates automatic selection of the USB mode where required, in Ao, A_2, A_3E and A_1, F_1, F_4 emissions.

f. Valid frequency indicator. An indicator shall be located immediately adjacent to the frequency select switch to denote that the frequency shown on this switch is consistent with that processed at the transmitter site. This indicator shall be labeled VALID, and shall be illuminated with a yellow lamp for a valid frequency comparison.

g. Sideband select validity. An indicator shall be located immediately adjacent to the sideband select switch to denote that the sideband shown on this switch is consistent with the selected class of emission. This indicator shall be labeled VALID and shall be illuminated with a yellow light when the sideband switch selection is valid for the selected class of emission.

The indicators specified in a through g shall be activated from decoded information extracted from the returned transmitter monitor messages.

3.4.1.9 RCU. One RCU shall be associated with each RCS. The RCU shall accept the parallel digital information from one to 10 DEUs and convert the information to serial digital data for transmission over a single pair of wires to the controlled transmitters via from one to eight local control multiplexers. In turn the RCU shall accept serial digital status data over a single pair of wires from the controlled transmitters via the local control multiplexer unit. The status data shall be converted to a parallel format and delivered to the DEU.

3.4.1.9.1 Status indicators. The RCU shall provide the indicators specified in a through e to display the control system status:

a. Monitor. The monitor indicator shall be illuminated with a white light to indicate that the RCS is monitoring the status of the controlled transmitters. The indicator shall be labeled MONITOR.

b. Control. The control indicator shall be illuminated with a white light to indicate that the RCS is commanding a controlled transmitter to a new state of operation. This indicator shall be labeled CONTROL.

c. Control fault. The control indicator shall be illuminated with a red light to indicate that a control command to a controlled transmitter could not be accomplished. This indication shall be labeled CONTROL FAULT.

d. Station monitor. Provisions shall be made for a split screen station monitor indicator for each local control multiplexer in the RCS. Four such indicators shall be provided and provisions shall be made to install four more as a field modification, if required by the RCS configuration. The upper portion of the indicator shall be internally illuminated by a yellow light and the lower portion shall be illuminated by a red light. The yellow portion shall light to indicate two successive errors in the data from a local control multiplexer. Eight more successive errors shall cause the red portion to become illuminated. Both portions of the indicator shall extinguish automatically upon receipt of error-free data. The yellow portion shall be labeled ERROR, and the red portion shall be labeled FAULT. The RCU shall be clearly marked to show that the four or eight indicators are STATION MONITOR indicators.

e. Audible alarm and reset. An audible alarm shall be provided that becomes activated when the control FAULT indicator is illuminated and when any station monitor fault indicator is illuminated. Latching circuitry shall be incorporated with the audible alarm to cause the alarm to continue to be activated after the station monitor FAULT indicator has extinguished. A three-position toggle switch shall be associated with the audible alarm so that the alarm may be disabled, or so that it may be reset to OFF after it has been activated. The reset position of the toggle switch shall be spring-loaded so that the switch will return to its center neutral position when the activating force is removed.

3.4.1.9.2 Variable word length. The RCU shall be capable of transmitting variable length control word messages of up to 40 characters, dependent upon the information required to control the end device. The control word transmission length shall be determined automatically as a function of operator or transmitter selection.

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3.4.1.10 Local control multiplexer. The local control multiplexer shall be located at the transmitter station. The local control multiplexer shall receive serial control information from the RCU and shall distribute this control information to the 10 possible local control interface units. The control message reply from the local control interface units shall be multiplexed by the local control multiplexer into a single serial bit stream for transmission back to the RCU.

3.4.1.10.1 Local control multiplexer to local control interface unit. Local control multiplexer to local control interface unit operation shall be as specified in 3.4.1.10.1.1 through 3.4.1.10.9.

3.4.1.10.1.1 Signalling. Signalling shall be as specified in a through d:

a. All control data from the local control multiplexer shall be transmitted over one pair of wires to the local control interface units. This line shall be terminated and driven by a differential line receiver and driver, respectively.

b. Transmitter operational status information from the local control interface unit to the local control multiplexer shall be binary coded and transmitted over three wires. The noise immunity application shall be not less than 2 volts (V), ±0.25 V, and shall have a hysteresis of not less than 400 millivolts.

c. Clock information from the local control multiplexer to the local control interface shall be transmitted over a single pair of wires. This line shall be terminated and driven by a differential line receiver and driver, respectively.

d. Up to four additional conductors can be utilized to transfer necessary control signals from the local control multiplexer to the local control interface unit. The line receivers used to terminate these four lines shall exhibit the same characteristics as specified in b.

3.4.1.10.2 Address decoding. Address decoding shall be as specified in a and b:

a. Station address. The local control multiplexer shall decode the station address character of each message received from the RCU. The multiplexer shall process the message and respond when the station address character coincides with the programmed address of the multiplexer. The multiplexer shall be made programmable to any address 0 to 9 by an internal switch.

b. Unit address. The local control multiplexer shall decode the unit address character of a control message when the station address character is in agreement with its programmed address. The unit address character information shall be used to perform the logic functions necessary for sending and receiving the control data to and from the proper local control interface unit.

3.4.1.10.3 Data error monitoring. Data error monitoring shall be as specified in a through c:

a. SOM. Once synchronization is established, the local control multiplexer unit shall decode the SOM character. If the SOM is not decoded properly, the remaining message shall not be processed. No reply will be initiated by the multiplexer unit which shall be detected by the RCU as a no-response.

b. All characters of a word transmission received by the multiplexer unit shall be checked to ensure containment of odd parity. A parity error occurring in address character, the system instruction, or the data instruction character shall inhibit the multiplexer from further processing of the message. Therefore, the multiplexer unit shall not respond to the control unit as a no-response.

c. If a parity error occurs in any of the remaining characters which specifiy class of emission, mode, or frequency, the multiplexer unit shall notify the addressed local control interface of this error occurrence. Notification of error shall inhibit the local control unit from processing the control parameter received in error. Only the parameter received in error shall not be processed. Those received without error shall be processed.

3.4.1.10.4 Data fault indications. Three successive data faults of any nature specified in 3.4.1.10.3 shall cause a fault alarm to sound and a data fault indicator to light on the multiplexer unit. These faults shall not be cumulative if they are not successive. The fault indicator shall automatically extinguish upon error-free reception of data by the multiplexer unit. A RESET/OFF switch shall be supplied for the audible alarm.

3.4.1.10.5 Local control interface unit synchronization. Synchronization for the local control interface unit shall be supplied by the local control multiplexer unit.

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3.4.1.10.6 Sensed end of receive. Correct timing for message word length shall normally be determined in the multiplexer unit by decoding the system instruction and unit address characters of the received message. If these characters are received in error and are not processed, the multiplexer unit shall be provided the ability to detect the end of message occurrence. This end of message sensing circuit shall activate in a time equal to five bit transitions after the end of message has occurred.

3.4.1.10.7 Variable word lengths. The local control multiplexer unit shall be capable of interfacing and controlling local control interface units which require various lengths of control messages, dependent upon the information required to control the end device. The multiplexer unit shall have the capacity to handle control words of up to 40 characters in length.

3.4.1.10.8 False start protection. Circuitry shall be provided in the local control multiplexer to prevent the signal decoding process from being initiated by a noise transient of 5 ms duration when operating at 75 baud.

3.4.1.10.9 Control separation. The RCS shall operate satisfactorily when the units are separated from each other with up to 100 meters (328.1 ft) of interconnecting cable.

3.4.1.11 Interface unit-local control. The local control interface unit shall be located in the space provided in the controlled transmitter. The interface unit shall convert the command data from the multiplexer to multiple wire binary signalling required for control of the transmitter via the MD-777/FRT Modulator-Synthesizer. The interface unit shall also accept the status signalling from the MD-777/FRT and shall convert the signalling to a format usable by the multiplexer.

3.4.1.11.1 Indicators. The local control interface unit shall contain the indicators specified in a through c to define the operating frequency and fault status of the associated transmitter.

a. Frequency. A frequency display shall provide a six-digit electronic readout indicating the operating frequency in MHz to the nearest hundred Hz of the associated transmitter. This display shall be blanked when the transmitter is set for local control operation.

b. Fault. The fault indicator shall be illuminated with a red light to indicate the receipt of a fault status signal from the associated transmitter or that a ready signal has not been received from the transmitter within a prescribed time interval. This indicator shall be labeled FAULT. The fault indication shall be cleared when the fault situation is corrected.

c. Audible alarm and reset. An audible alarm that becomes activated when the FAULT indicator is illuminated shall be provided. A three-position toggle switch shall be associated with the audible alarm so that the alarm may be disabled, or so that the switch may be reset off after it has been activated. The reset position shall be spring-loaded so that the switch will return to its center neutral position when the activating force is removed.

3.4.1.11.2 Tuning time monitor. A circuit that will generate a fault signal when the ready read-back signal from the associated transmitter is not received within a specified time interval from the time the tune command is given to the transmitter shall be incorporated into the interface unit. Time intervals of 15 seconds, 30 seconds, or 60 seconds shall be selectable by jumper wires located within the interface unit.

3.4.1.11.3 Line filters. All interface wiring from the interface unit to the associated transmitter and to the multiplexer shall enter the interface unit through line filters. The filters on the wiring between the interface unit and the transmitter shall provide a capacitance to chassis ground of not less that 50,000 picofarads (pF) nor more than 100,000 pF. The filters on the signal wiring to the multiplexer shall provide a capacitance to ground of not less than 10,000 pF nor more than 20,000 pF.

3.4.1.11.4 Transmitter signalling. The transmitter signalling from the interface unit shall be via solid-state circuitry which has a minimum break-down voltage of 40 VDC and a minimum current sinking capability of 40 mA. The presence of a metallic closure on a particular read-back wire to its respective common wire, as specified in TABLE VI shall indicate that the transmitter is functioning in that particular operational condition. The read-back wires from the transmitter shall be connected within the interface unit to a positive voltage of 5 VDC ±10 percent through a resistance of 620 ohms ±20 percent. The common wires of the read-back signals shall be connected within the interface unit to the negative side of the positive 5 VDC power source.

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3.4.1.11.5 Transmitter read-back. Rotary mechanical switching is used in the MD-777/FRT Exciter of the controlled transmitter for sideband selection and class of emission selection, and these switches may not have reached their commanded position before a reply message is generated in the local control interface unit. Therefore, read-back information for message comparison of these parameters shall be taken from the storage register within the local control interface unit. To ensure that these switches settle on their commanded position, a comparison shall be made in the local control interface unit of the read-back from the transmitter and the coding contained within the interface unit storage. Failure of either switch to settle on its commanded position within the tune-time frame specified in 3.4.1.11.2 shall be recorded as a transmitter fault at the DEU.

3.4.1.11.6 Transmitter interface connectors. The interface connection to the transmitter shall be through the two connectors specified in TABLE VI. These two connectors shall be located on the rear panel of the interface unit. The exact location of the connectors on the interface unit shall be consistent with GFE equipment (see 3.2.1.1).

3.4.2 Operation. The RCU of the remote control group shall normally operate in the monitor mode. In the monitor mode the control unit shall sequentially interrogate each local control multiplexer in the remote system to obtain operational status of all transmitters assigned to the multiplexers. This monitor mode of operation shall be interrupted by operator selection of the control or status modes. Upon completion of the control or status mode, the RCU shall revert to the monitor mode of operation. The monitor interrogation sequence shall not be disrupted as result of a control or status mode interrupt.

3.4.2.1 Fault notification. A transmitter fault notification during the monitor mode shall cause the appropriate tranmitter fault indicator to illuminate and the fault alarm to sound. A fault notification from any transmitter shall not disrupt the remaining monitor sequence processing. A redundant transmitter fault notification shall not cause the audible alarm to activate. Disconnection of the signal cable between the local control multiplexer and a particular local control interface unit shall be detected as a transmitter fault for the associated transmitter.

3.4.2.2 Failure of data channel. A control unit failure or loss of the data channel between the RCU and local control multiplexer shall not disrupt transmitter operations.

3.4.2.3 Status display. The control mode of operation shall be terminated upon proper validation of the returned transmitter data (see 3.4.1.3.3) or upon two transmissions of the control message as specified in 3.4.1.3.5. After completion of a properly validated control message, the DEU shall display the current transmitter operational status (tune, fault, or operate and ready).

3.4.2.4 Priority. In systems which incorporate more than one DEU, which allows for more than one control request to be present, no two successive and different control messages shall be transmitted without being alternated with a monitor mode interruption. In addition, control requests from the DEUs shall be honored sequentially by the control unit on a priority basis if more than one request is entered.

3.4.3 Physical characteristics. Physical characteristics shall be as specified in 3.4.3.1 through 3.4.3.4.

3.4.3.1 DEU. The DEU shall be 1.7 in. high, 14 in. deep, 17.5 in. wide, and shall be equipped with mounting flanges for installing in standard 19-in. racks. The DEU shall not weigh more than 20 pounds (lbs).

3.4.3.2 RCU. The RCU shall be 5.25 in. high, 18 in. deep, 17.5 in. wide, and shall be equipped with mounting flanges for installing in standard 19-in. racks. The RCU shall not weigh more than 40 lbs.

3.4.3.3 Local control multiplexer. The multiplexer shall be 7 in. high, 17.5 in deep, 17.5 in. wide, and shall be equipped with flanges for installing in standard racks. Entrance to this unit for servicing and maintenance shall be through a hinged front panel. Opening the front panel shall expose all of the panels PCCs, PCC connectors, and shall present ready access to all modules of the unit. This unit shall not weigh more than 50 lbs.

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3.4.3.4 Local control interface unit. The interface unit shall be 7 in. high, 17.5 in. deep, 17.5 in. wide, and shall be equipped with flanges for installing directly under the MD-777 Modulator-Synthesizer in the associated transmitter. Entrance to this unit for servicing and maintenance shall be through a hinged front panel. Opening the front panel shall expose all of the panel's PCCs, PCC connectors, and present ready access to all of the modules of the unit. This unit shall not weigh more than 40 lbs.

3.5 Safety. The safety features of the RCS shall be in accordance with the provisions of MIL-STD-454, Requirement 1.

3.6 Connectors. The contractor shall provide all necessary mating connectors for interconnecting cables for the RCS.

3.7 Quantitative reliability requirements. The upper test MTBF (θ_0 as defined by MIL-STD-781) shall be 5000 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 2500 hours. The MTBFs of the AN/FRT-84( )(600 hours) and associated AN/FSQ-98(V)(2500 hours) are mutually exclusive, that is, a failure in one equipment shall not be treated as a failure in the other.

3.8 Workmanship. Workmanship shall be as specified in 3.8.1.

3.8.1 General workmanship. Workmanship and soldering shall conform to the Workmanship and Soldering paragraphs of MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (Group A) (see 4.4.1)
    2. Production control inspection (Group B) (see 4.4.2)
    3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2), two sample units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE VIII. First article inspection shall be performed while the RCS is configured and installed in a Radio Transmitter, AN/FRT-84( ), with which the RCS is intended to be used.

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TABLE VIII. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Fungus	3.1	4.5.1	X			
Humidity	3.1, 3.2.10	4.5.1	X			X
Dust	3.1, 3.2.10.3	4.5.2	X			
Temperature (operating and nonoperating)	3.2.10	4.5.4	X			X
Shock	3.2.3	4.5.3	X			X
Elevation	3.2.10	4.5.7	X			X
Airborne and structureborne noise	3.1	4.5.1	X			
Enclosure	3.1, 3.2.2	4.5.1	X			
Weld	3.1	4.5.1	X			
Steady-state voltage and frequency	3.1	4.5.1	X			
Transient voltage	3.1	4.5.1	X			
Transient frequency	3.1	4.5.1	X			
Spike voltage	3.1	4.5.1	X			
Power interruption	3.1	4.5.1	X			
Power and power factor	3.1	4.5.1	X			
Insulation resistance	3.1	4.5.1	X			
Dielectric withstanding voltage	3.1	4.5.1	X			
Electromagnetic interference	3.2.8	4.5.8	X			X
Electromagnetic field radiated susceptibility test	3.2.8	4.5.8	X			X
Thermal design	3.1, 3.2.14	4.5.4	X			
Accelerated life	3.1	4.5.1	X			
Burn-in	3.1.2	4.5.6	X		X	
Reliability	3.7	4.6	X			
General workmanship	3.8.1	4.7.1	X	X		
DEU						
Control functions	3.4.1.8.1	4.5	X	X		
Status indicators	3.4.1.8.2	4.5	X	X		
RCU						
Status indicators	3.4.1.9.1	4.5	X	X		
Variable word lehgth	3.4.1.9.2	4.5	X	X		
Mode test switch	3.2.18	4.5	X	X		
Local control multiplexer, TD-1102/FSQ-98(V)						
Local control multiplexer to local control interface unit	3.4.1.10.1	4.5	X	X		
Address decoding	3.4.1.10.2	4.5	X	X		
Data error monitoring	3.4.1.10.3	4.5	X	X		
Data fault indications	3.4.1.10.4	4.5	X	X		
Sensed end of receive	3.4.1.10.6	4.5	X	X		
Variable word length	3.4.1.10.7	4.5	X	X		
Mode test switch	3.2.18	4.5	X	X		
Local control interface unit, J-3200/FSQ-98(V)						
Indicators	3.4.1.11.1	4.5				
Tuning time monitor	3.4.1.11.2	4.5	X	X		
Transmitter signalling	3.4.1.11.4	4.5	X	X		
Transmitter read-back	3.4.1.11.5	4.5	X	X		
RCS control signal electrical characteristics						
Rate	3.4.1.1.2a	4.5	X	X		
Levels	3.4.1.1.2c	4.5	X	X		

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TABLE VIII. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Signal power supply	3.4.1.1.2d	4.5	X	X		
System distortion	3.4.1.2.2	4.5	X	X		
Data link error control and monitoring						
Parity	3.4.1.3.1	4.5	X	X		
Address comparison	3.4.1.3.2	4.5	X	X		
Message comparison	3.4.1.3.3	4.5	X	X		
Channel monitoring	3.4.1.3.4	4.5	X	X		
ARQ	3.4.1.3.5	4.5	X	X		
Fault notification	3.4.2.1	4.5	X	X		
Test cards and module extenders	3.2.13	4.5	X		X	
Interchangeability	3.2.17	4.5	X		X	
I/O characteristics 1/	3.4.1.1.3	4.5 2/	X		X	
Message word format 3/ and encoding and language	3.4.1.1.2, 3.4.1.1.4, 3.4.1.5	4.5 2/ 4.5	X		X	
Synchronization	3.4.1.4	4.5	X		X	
Propagation delays	3.4.1.6	4.5	X		X	
Polling time requirements	3.4.1.6.1	4.5	X		X	
Primary power	3.4.1.7	4.5.5	X		X	
RCU						
Loss of power	3.4.1.7.4	4.5.5	X		X	
Local control multiplexer, TD-1102/FSQ-98(V)						
False start protection	3.4.1.10.8	4.5	X		X	
Control separation	3.4.1.10.9	4.5	X		X	
Local control interface unit, J-3200/FSQ-98(V)						
Line filters	3.4.1.11.3	4.5	X		X	
Failure of data channel	3.4.2.2	4.5	X		X	
Status display	3.4.2.3	4.5	X		X	
Priority	3.4.2.4	4.5	X		X	
Physical characteristics						
DEU	3.4.3.1	4.5	X		X	
RCU	3.4.3.2	4.5	X		X	
Local control multiplexer, TD-1102/FSQ 98(V)	3.4.3.3	4.5	X		X	
Local control interface unit, J-3200/FSQ-98(V)	3.4.3.4	4.5	X		X	

1/ See TABLES III, IV, V, and VI.
2/ As production control inspections, these tests shall be performed on two systems from the first inspection lot and need not be reperformed on subsequent lots.
3/ See TABLE II.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.3.2.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on every RCS offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the examinations and tests specified in TABLE VIII.

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4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis in accordance with the inspection procedures of MIL-STD-105 to an acceptable quality level (AQL) of 6.5 percent using the special inspection level S-3 for normal, tightened, and reduced inspection. Production control inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of sets to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily conform to the requirements of production control inspection prior to release for shipment. Production control inspection shall include the examination and tests shown in Group B of TABLE VIII.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be performed on RCSs which have been subjected to, and have passed, production inspection. Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests, simulated service tests; test of the effects of changes of environment (such as extremes of temperature and humidity, effect of salt air); and tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examination and tests shown in Group C of TABLE VIII.

4.4.3.1 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and details of action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.5).

4.4.3.2 Reinspection of conforming environmental sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to, and passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Test methods. Unless otherwise specified, the RCS shall be subjected to performance tests to determine compliance with the specified performance parameters. The test methods shall be determined by the contractor subject to Government approval.

4.5.1 Examinations and tests. Examinations and tests specified in TABLE VIII shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.5.2 through 4.5.9.

4.5.2 Dust. During the first article inspection, the sample units shall be tested in accordance with MIL-STD-810, Method 510.2, Procedure I, except that testing in the operating condition is excluded, and the ambient temperature shall not exceed 50°C.

4.5.3 Shock. Shock testing shall be in accordance with MIL-STD-810, Method 516.3, Procedure I, and the Incline impact test specified in the Rough handling test table of MIL-P-116.

4.5.4 Temperature. The temperature test shall be performed in accordance with MIL-E-16400, Temperature Range 4.

4.5.5 Primary power. The RCS shall be tested as specified in TABLE IX to determine conformance with 3.4.1.7. Reference performance measurements shall be made at 115 VAC and 60 Hz. Performance measurements shall then be made for each condition specified in TABLE IX after the RCS has operated for 15 minutes in each condition.

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TABLE IX. Input voltage and frequency.

Condition	Voltage (VAC)	Frequency (Hz)
1	103.5	47
2	126.5	47
3	126.5	63
4	103.5	63
5 1/	207.0	47
6 1/	253.0	47
7 1/	253.0	63
8 1/	207.0	63

1/ As production control inspections, these test conditions need not be reperformed after these tests have been performed on a minimum of two complete systems.

4.5.6 Burn-in. Each RCS shall be subjected to a 48-hour burn-in. Should a failure occur within the last 24 hours of the test, appropriate corrective action shall be taken and the equipment shall be operated until 24 continuous hours of failure-free operation have accrued.

4.5.7 Elevation test. Prior to beginning the elevation tests, the RCS shall be given a visual examination and electrical tests to determine conformance to all production inspections. The steps specified in a through d shall be performed to determine conformance to 3.2.10:

a. Place the RCS in the altitude chamber and allow the unit to stabilize at normal room ambient temperature. The temperature shall be maintained at normal room ambient during the entire test.

b. Reduce the chamber internal pressure to 525 millimeters of mercury (10,000 ft above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for not less than 1 hour.

c. Return the chamber to room ambient conditions.

d. The RCS shall be subjected to production inspection to determine that no degradation occurred as a result of the altitude test. The inspection shall include a visual examination to determine that no deterioration resulted from the test.

4.5.8 Electromagnetic capabilities characteristics. The RSC shall be tested to confirm conformance to the requirements of 3.2.8. Test RS03 of MIL-STD-462 is not required for this testing.

4.5.9 Operating test. Each DEU, ID-1925/FSQ-98(V); RCU, C-9427/FSQ-98(V); Local Control Multiplexer, TD-1102/FSQ-98(V); and Local Control Interface Unit, J-3200/FSQ-98(V), shall be assembled into a system configuration and subjected to an operating test to ensure qualitatively the correct functioning of the RCS. All unused outputs of the local control multiplexer shall be terminated into the proper load impedance to simulate the local control interface units in the RCS. A minimum of five local control multiplexer outputs shall be employed for this test. A minimum of five station addresses shall be randomly selected at each local control multiplexer employed in the RCS with the corresponding address changes made at the remote end. As a minimum, the functions specified in a through d shall be performed using the GFE AN/FRT-84 Radio Transmitter.

a. Random frequency selection of at least five frequencies with a minimum separation of 2 MHz

b. Use of all operating modes

c. Simulation of data link interruption

d. Simulated failure of GFE transmitter with confirmation of fault

4.6 Reliability. Reliability shall be as specified in 4.6.1 through 4.6.4.

4.6.1 Reliability demonstration. A reliability demonstration shall be performed on at least one unit in accordance with Test Plan XXC of MIL-STD-781 to demonstrate the equipment upper and lower test MTBF specified in 3.7. Environmental and operational duty cycles shall be as specified in 4.6.1.1 through 4.6.1.5.

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4.6.1.1 Temperature cycling. Temperature cycling shall be performed between 0°C and +50°C while the RCS is operating.

4.6.1.2 Input-voltage cycling. Input voltage cycling shall be performed in accordance with TABLE IX.

4.6.1.3 Humidity cycling. Humidity cycling shall be performed between 0 percent and 95 percent RH while the RCS is operating.

4.6.1.4 Operation duty cycle. The RCS shall operate continuously until the end of the reliability demonstration.

4.6.1.5 Performance measurements. At least daily, baud rate measurements shall be made to ensure proper RCS operation.

4.6.2 Failure. The procuring activity shall be notified of any failure during the reliability demonstration. Classification of failure shall be made by the procuring activity in accordance with the Failure categories paragraph of MIL-STD-781.

4.6.3 Rejected equipment. If equipment is rejected as a result of reliability testing, the contractor shall immediately notify the procuring activity of the cause(s) for rejection and propose a plan for correction of the deficiencies. When such corrections have been approved by the procuring activity and incorporated in the unit, the reliability demonstration shall continue as specified herein.

4.6.4 Disposition of test equipment. Upon successful completion of the reliability test, the equipment shall be returned to the production area for refurbishment as necessary and shall be submitted to the production inspection of 4.4.1. The equipment may be delivered as part of the contract quantity, provided the equipment conforms to all the requirements of the contract.

4.7 Workmanship. Workmanship shall be as specified in 4.7.1.

4.7.1 General workmanship. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements of 3.8.1. Each soldering connection and associated lead shall be visually examined.

4.8 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. This specification covers a solid-state RCS to be used in conjunction with the AN/FRT-83(V), AN/FRT-84(V), and AN/FRT-85(V), High-Frequency Radio Transmitters in Navy shore installations throughout the world.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than as specified in 3.1.1 and 4.3
c. Adjustment of the production control inspection sampling plan, as required, for compatibility with end item delivery requirements (see 4.4.2)
d. When reinspected environmental sample units may not be accepted (see 4.4.3.2)
e. Levels of preservation, packaging, packing, and marking required (see 5.1)

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6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 52.209-3 of the Federal Acquisition Regulations. The first article should be a first production item. The first article should consist of two units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

6.5 Initial shipment. Initial shipment of end items will be contingent upon successful completion of the environmental tests. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.6 Synonyms used for major components within this specification. Synonyms used for major components within this specification are specified in a through c:

Official term	Synonymous term
a. Interface unit-local control, J-3200/FSQ-98(V)	Interface unit Local control interface Local control interface unit Local control unit
b. Multiplexer-Demultiplexer, TD-1102/FSQ-98(V)	Local control multiplexer Multiplexer Multiplexer unit
c. Remote control group	Control unit
1. Control Unit, Remote C-9427/ FSQ-98(V)	Control and entry units DEU
2. Display and Entry Unit, ID-1925 FSQ-98(V)	Display unit DEU

6.7 GFE. The contracting officer should arrange to furnish the GFE listed in 3.2.16 and 4.5.9 (see 3.2.1.1).

Preparing activity:
NAVY-EC

(Project 5820-N779)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-28792A	Control System, Remote, AN/FSQ-98(A)(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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